

TOUQEER AHMAD

Data Scientist | Applied ML, NLP & Production AI Systems

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PROFESSIONAL SUMMARY

Data Scientist and Machine Learning practitioner with hands-on experience across the AI stack from unsupervised learning on large-scale real world data (750K+ records) to fine-tuning and deploying transformer based language models. Currently pursuing an M.Sc. in Data Science & AI at BITS Pilani. Distinguished by the ability to take models beyond the notebook: independently designs, fine tunes, evaluates, and ships production grade AI and AWS systems including a LoRA fine-tuned NLP translation model and a deployed retrieval augmented generation (RAG) application giving equal fluency in model development and the engineering required to deploy it. Civil engineering background adds a grounded, domain driven approach to applied AI problems.

TECHNICAL SKILLS

Programming & Data: Python (Pandas, NumPy, Scikit-learn), SQL, PostgreSQL, JavaScript/Node.js

Machine Learning: K-Means, Hierarchical (Ward/Average Linkage) & DBSCAN Clustering, Linear Regression, Random Forest, XGBoost, Feature Engineering, Silhouette Analysis, ANNs

NLP & LLM Systems: Transformer Fine-Tuning (LoRA / PEFT), Hugging Face Transformers, Retrieval-Augmented Generation (RAG), LangChain, Vector Databases (ChromaDB), Sentence Embeddings, LLM APIs (Anthropic Claude, Google Gemini), Prompt Engineering, Model Evaluation (BLEU, chrF)

Deep Learning & Deployment: PyTorch, Hugging Face Hub & Spaces, Gradio, Model Deployment & Inference Optimization

Data Engineering: Kafka, PySpark, ETL Pipelines, Docker, Zookeeper

Analytics & Visualization: Excel (Advanced), Tableau, Matplotlib, Customer/Behavioral Segmentation

Cloud & Infrastructure: AWS (Lambda, DynamoDB, API Gateway, S3, CloudFront, Route 53, SES, SNS, ACM), Oracle Cloud Infrastructure, Oracle Cloud Generative AI

Software Engineering: Full-Stack Web Development, REST APIs, Serverless Architecture, Git/GitHub, CI/CD-style Deployment Automation

KEY PROJECTS

Kashmiri-English Neural Machine Translation Model | Personal Project — 2026

- Fine-tuned Meta's NLLB-200 (600M-parameter) multilingual translation model using **LoRA (PEFT)** on 5,573 English Kashmiri sentence pairs, addressing a critically underserved low resource language.
- Designed a rigorous evaluation pipeline (**BLEU, chrF**) against a held-out test set, quantifying real gains over baseline (BLEU +0.37, chrF +0.63) and documenting results transparently in a published model card.
- Diagnosed and resolved multiple production ML issues independently: a padding token masking bug causing training instability, a floating point precision loss from FP16 weight merging, and a Hugging Face Spaces ZeroGPU deployment conflict.
- Published fine tuned model and adapter weights on Hugging Face Hub with full documentation; deployed an interactive public demo (Gradio) on Hugging Face Spaces.

Recruiter-Facing RAG Chatbot | Personal Project — 2026

- Designed and deployed a **Retrieval-Augmented Generation** system that answers questions about my background using my own CV as a grounded knowledge base, with source citations on every answer built on LangChain, ChromaDB, and sentence transformers embeddings, with a swappable LLM backend (Claude / Gemini).
- Engineered an automated content-sync pipeline: pushing an updated CV or project document to GitHub triggers automatic re-indexing on the live deployment via a content-hash-based rebuild check, requiring no manual redeployment steps.
- Deployed live on Streamlit Community Cloud with secrets-based configuration management, demonstrating end to end ownership from model design through production deployment.

Customer Behaviour Segmentation on Banking Transaction Data | M.Sc. Apex Capstone Project — 2025

- Applied K-Means, Hierarchical Clustering, and DBSCAN to segment ~753K bank customers by transaction behavior, handling a heavily skewed dataset where 86.28% of customers had only a single transaction.
- Tuned and validated across techniques: K-Means (k=3) achieved a silhouette score of 0.3317 with balanced cluster sizes; identified severe chaining in average-linkage hierarchical clustering and established Ward linkage as the more suitable alternative; optimized DBSCAN to eps=1.009, min_samples=3.
- Delivered as a fully reproducible, runnable Colab notebook, correctly scoping evaluation methods to the unsupervised setting.

House Price Prediction Model | M.Sc. Coursework, BITS Pilani — 2026

- Trained and evaluated Linear Regression, Random Forest, and XGBoost models on Bengaluru housing data, applying feature engineering and hyperparameter tuning to improve predictive accuracy.

Streaming Data Ingestion Pipeline | M.Sc. Coursework, BITS Pilani — 2026

- Built a real-time streaming ingestion pipeline using Kafka, PySpark, PostgreSQL, Zookeeper, and Docker to process and land streaming data end-to-end.

Patient Booking & Authentication System | Freelance | drtejasvisharma.in — 2025 – Present

- Designed a full email-based OTP authentication and appointment booking system on AWS, and an SES bounce/complaint-handling pipeline (SNS → Lambda) that automatically flags suppressed patient emails in DynamoDB direct experience turning a data signal into an automated, production decision.

Shaggykitchen — Serverless Food Ordering Platform | Owner & Developer | shaggykitchen.in — Feb 2026 – Present

- Shipped a production-grade serverless ordering platform (Customer, Owner, Delivery Agent roles) on AWS; built an Owner Analytics dashboard with churn tagging and menu performance metrics, and map-based delivery-fee logic using the Haversine formula end-to-end ownership from data to decision to deployed feature.

Ecommerce Seller Recommendation System | Personal Project — 2025

- Built ETL pipelines across seller catalog, company sales, and competitor sales data in Python to recommend top selling items.

Data Analytics & Data Science Job Simulations | Forage — Deloitte Australia · Commonwealth Bank — Feb 2026

- Completed real world analytics tasks spanning Tableau dashboarding, forensic data analysis, exploratory data analysis, feature identification, and stakeholder communication of findings.

PROFESSIONAL EXPERIENCE

Data-Driven Client Advisory (Insurance) | Star Health and Allied Insurance / LIC — Aug 2023 – Present

- Segment client demographics and financial profiles using Excel-based analysis to tailor product recommendations and prioritize outreach applied, informal analytics experience predating formal ML training.
- Track and analyze sales pipeline performance to identify conversion trends and inform retention strategy.

EDUCATION

M.Sc. in Data Science & Artificial Intelligence: BITS Pilani (Sep 2025 – Sep 2027, Expected)

B.E., Civil Engineering: Model Institute of Engineering and Technology (MIET), Jammu (2019 – 2022)

Diploma in Civil Engineering (3-Year): Government Polytechnic College, Jammu (Sep 2016 – Jul 2019)

CERTIFICATIONS & TRAINING

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional — Oracle (Oct 2025)
- Python Program: Coursera / UST (Sep 2025)
- Data Visualization: Coursera (Sep 2025)
- Excel & Master Data Analytics in Excel: Coursera / Great Learning (Sep 2025)
- Practical SQL Training: Great Learning (Sep 2025)

PROFESSIONAL MEMBERSHIPS

- Student Member, American Society of Civil Engineers (ASCE) — since Jul 2020